

Custom QGroundControl Test Manual

A practical flight-test checklist for validating the custom QGroundControl build from the local **devel** branch against the expected pilot-facing behavior.

BRANCH UNDER TEST

devel

BASELINE

master

PREPARED FOR

Test pilot

REVISION DATE

2026-05-12

1. Purpose And Scope

This manual tells the pilot what to test, how to test it, and what result is expected. It focuses on the custom changes visible to pilots: HUD, FPV/Stribog interface selection, camera and gimbal behavior, terrain and altitude display, toolbar indicators, and basic operational safety.

Branch note: local `devel` was reported as 5 commits behind `origin/devel`, and `master` also contains commits not present in `devel`. Record the exact application version and commit hash before testing.

Main Areas To Validate

FLIGHT UI

Custom HUD, FPV/Stribog layout, toolbar indicators, compass, altitude and terrain display.

CAMERA AND GIMBAL

T20/SIYI gimbal slider, multi-camera FOV, camera discovery, video startup and stream switching.

MISSION AND SAFETY

Map click behavior, guided controls, route display, telemetry, link status, battery status and fail-safe visibility.

2. Safety Rules

Do not begin flight testing until bench testing passes. Any crash, frozen UI, missing telemetry, wrong flight mode, wrong altitude, wrong heading, uncontrolled gimbal motion, or wrong camera feed is a stop condition.

- ☐ Use a safe test area with legal flight permission, weather limits, and clear emergency landing space.
- ☐ Keep a manual joystick ready and verified before arming.
- ☐ Use a visual observer when testing camera, FPV, or HUD changes.
- ☐ Start with props removed for bench tests, then low-altitude hover, then normal mission tests.
- ☐ Record screen video or logs for every failed test.
- ☐ Stop testing immediately if QGroundControl displays inconsistent state compared with the vehicle.

3. Test Setup

Required Equipment

Item	Purpose	Record Value
Custom QGroundControl build	Application under test	Version / commit hash:
Vehicle	Flight platform for validation	Model / firmware:
Camera payload	Video, FOV and gimbal validation	Camera type:
Gimbal	T20/SIYI slider and control validation	Gimbal type:
Ground station device	Display and control device	OS / screen size:
Telemetry link	Vehicle data and command link	Radio / network type:

Before Each Test Session

- ☐ Confirm application starts without crash or blocking error dialog.
- ☐ Confirm correct build is installed and the version popup or about window shows the expected version/hash.
- ☐ Confirm vehicle connects and shows correct vehicle type, flight mode, battery, GPS and link state.
- ☐ Confirm map tiles or offline map data are available for the test area.
- ☐ Confirm video source and gimbal payload are powered and connected.
- ☐ Start screen recording if available.

Pilot UI Setup Click Paths

Task	What To Click	Expected Result
Show the AGL terrain UI element	Open Fly View . In the bottom-right instrument panel, right-click the instrument on desktop, or long-press it on a touch device. Open the selector and choose AGL . Click the lock/open button to close the selector.	The bottom-right instrument changes to the AGL terrain profile and shows height above ground/terrain information when terrain data is available.
Force FPV layout manually	Open Application Settings using the gear icon. Select General . Find UI Layout . Select FPV .	The Fly View changes to the FPV layout with the custom edge controls visible.
Use Stribog / normal layout	Open Application Settings using the gear icon. Select General . Find UI Layout . Select	The Fly View returns to the non-FPV layout used for the standard/Stribog pilot

Task	What To Click	Expected Result
manually	Normal . If this build labels the option as Stribog instead of Normal , select Stribog .	workflow.
Use automatic layout switching	Open Application Settings using the gear icon. Select General . Find UI Layout . Select Auto , then connect the vehicle.	QGroundControl selects the custom FPV layout only when the connected vehicle metadata identifies the vehicle as an FPV model; otherwise it stays in the normal/Stribog layout.

4. Test Matrix

ID	Area	Priority	When To Run	Pass Criteria
T01	Install and launch	Critical	Every build	App opens, version is visible, no startup crash.
T02	Vehicle connection and telemetry	Critical	Bench and flight	Vehicle state matches real vehicle and updates continuously.
T03	Automatic UI switching	High	Bench	Correct FPV/Stribog UI appears for selected or detected model.
T04	Custom HUD	Critical	Bench and flight	Heading, altitude, terrain and attitude are readable and correct.
T05	Compass and azimuth	High	Bench and flight	Compass direction and azimuth values are stable and plausible.
T06	Terrain and AGL indication	High	Bench and low flight	Terrain indicator loads data and does not show misleading values.
T07	Camera discovery and video	Critical	Bench and flight	Correct camera feed appears and recovers after reconnect.
T08	Multi-camera FOV	High	Bench and flight	FOV changes match selected camera and gimbal orientation.
T09	T20/SIYI gimbal slider	Critical	Bench before flight	Slider motion is smooth, bounded and matches gimbal movement.
T10	Toolbar indicators	High	Bench and flight	Battery, flight mode, link, GPS and remote state are visible and correct.
T11	Map interaction	Normal	Bench and flight	Detached map clicks and zoom actions do not trigger wrong actions.
T12	Regression smoke test	Critical	Before release	Arm, mode change, takeoff, land, video, logs and settings all work.

5. Test Procedures

T01 - Install, Launch And Version Check

Critical

Goal: Confirm the pilot is testing the correct custom build.

1. Install or start the custom QGroundControl build.
2. Open the version/about information or version popup.
3. Record the branch, version, build date and commit hash if shown.
4. Close and reopen the application.

Expected result: QGroundControl starts cleanly every time. The version information is visible and identifies the intended custom build.

Fail if: the app crashes, hangs, shows missing dependency errors, or version/hash cannot be confirmed.

T02 - Vehicle Connection And Telemetry

Critical

Goal: Confirm QGroundControl correctly detects and tracks the vehicle.

1. Connect telemetry with the vehicle disarmed.
2. Verify vehicle type, firmware type, flight mode, armed state, battery and GPS status.
3. Move or rotate the vehicle by hand during bench test.
4. Disconnect and reconnect the telemetry link.

Expected result: values update continuously, reconnect works, and no stale armed/mode/battery/GPS state remains after reconnect.

Flight check: during hover, compare displayed altitude, heading and mode with the RC/controller or autopilot source.

T03 - Automatic FPV / Stribog UI Switching

High

Goal: Validate the custom UI selection behavior from the `feat/initial_auto_switch` and `unify/fpv_stribog_unify` changes.

1. Start QGroundControl with no vehicle connected and note the default UI.
2. Connect each supported test vehicle/model one at a time.
3. Observe whether the correct FPV or Stribog layout appears automatically.
4. Open **Application Settings** with the gear icon, select **General**, then use **UI Layout** to test **Normal**, **Auto**, and **FPV**.
5. If the non-FPV option is labeled **Stribog** on the tested build, use **Stribog** where this manual says **Normal**.
6. Return **UI Layout** to the expected release setting before flight, normally **Auto** unless the test plan says otherwise.

Expected result: the correct UI appears for the selected or detected vehicle model. Manual selection persists after restart if that is intended.

Fail if: the wrong UI appears, switching requires app restart unexpectedly, or controls disappear after model selection.

T04 - Custom HUD Readability And Correctness

Critical

Goal: Validate custom HUD, compass, altitude and terrain presentation.

1. Open Fly View with video disabled, then enabled.
2. Check that HUD text and indicators are readable on the ground station screen in sunlight or expected field lighting.
3. Pitch, roll and yaw the vehicle by hand with props removed.
4. During low hover, check attitude, altitude, vertical movement and heading behavior.
5. Change map/video layout if available and confirm the HUD remains readable.

Expected result: HUD elements do not overlap, flicker, vanish or cover critical controls. Heading, altitude and attitude match vehicle movement.

Fail if: the HUD shows misleading values, unreadable text, unstable scaling, clipped labels, or wrong orientation.

T05 - Compass, Azimuth And Heading

High

Goal: Confirm the compass resize, scaling and azimuth fixes behave correctly.

1. Place the vehicle facing north, east, south and west if practical.
2. Compare QGroundControl heading with an independent compass or autopilot status.
3. Rotate slowly through 360 degrees and watch for jumps or wraparound errors.
4. Repeat once with the video/HUD overlay active.

Expected result: heading changes smoothly, numeric azimuth is readable, and the compass stays correctly sized.

T06 - Terrain And AGL Indicator

High

Goal: Validate offline terrain elevation and dynamic terrain indicator changes.

1. Load the map for the test area before flight.
2. Open **Fly View**.
3. In the bottom-right instrument panel, **right-click** the instrument on desktop, or **long-press** it on a touch device.
4. Open the selector and choose **AGL**. Click the lock/open button to close the selector.
5. Confirm the AGL terrain indicator appears in the bottom-right instrument area.
6. Test with network available, then with network disabled if offline terrain is expected.
7. During low flight, compare AGL trend with expected terrain changes.

Expected result: terrain data loads without freezing the UI. The indicator clearly shows unknown/unavailable data instead of a misleading value.

Fail if: terrain values jump unrealistically, remain stuck, or make the pilot believe the vehicle is safer than it is.

T07 - Camera Discovery, Video And Reconnect

Critical

Goal: Confirm camera discovery and video display are stable.

1. Start QGroundControl before powering the camera payload.
2. Power the payload and wait for camera discovery.
3. Confirm video starts on the expected camera feed.
4. Switch cameras or streams if multiple feeds are available.
5. Power-cycle the payload or disconnect/reconnect the stream.

Expected result: video starts, selected feed is correct, reconnect works, and the app does not crash on camera loss.

Fail if: video is black without a clear state, wrong feed is selected, reconnect requires full application restart, or controls become unresponsive.

T08 - Multi-Camera FOV And Gimbal FOV

High

Goal: Validate the `feat/auto_fov` changes for multi-camera FOV handling.

1. Select each available camera feed.
2. Observe FOV overlay, framing or map/video FOV indicator if available.
3. Move the gimbal through pan and tilt ranges.
4. Zoom or switch lens/FOV mode if supported by the payload.

Expected result: FOV updates correspond to the active camera and gimbal direction. Switching cameras does not leave stale FOV data from a previous camera.

T09 - T20 / SIYI Gimbal Slider

Critical

Goal: Validate gimbal slider movement and axis guard behavior.

1. Run this test on the bench before any flight.
2. Move the slider slowly from center to each limit.
3. Release the slider and confirm the gimbal stops as expected.
4. Move the slider quickly and check for delayed or excessive motion.
5. Repeat while video is active and while telemetry quality is reduced, if safe to simulate.

Expected result: the gimbal moves in the correct direction, respects limits, stops reliably, and does not command the wrong axis.

Fail if: the gimbal moves unexpectedly, oscillates, ignores limits, uses the wrong axis, or keeps moving after control release.

T10 - Toolbar Indicators

High

Goal: Confirm new or moved toolbar indicators remain clear and correct.

1. Check battery, flight mode, armed state, GPS, telemetry RSSI, link and gimbal indicators.
2. Open each indicator panel and close it again.
3. Change flight mode from the normal pilot workflow.
4. Simulate weak GPS or telemetry loss if safe and controlled.

Expected result: indicators show correct state, panels fit on screen, and controls do not block important flight information.

T11 - Map Interaction And Detached Map Clicks

Normal

Goal: Confirm map clicks and layout changes do not cause unintended actions.

1. Open Fly View and interact with the map using click, drag and zoom.
2. Switch between video-focused and map-focused layouts if available.
3. Open and close guided action menus.
4. Verify clicking detached map areas does not trigger wrong buttons or commands.

Expected result: map interactions are predictable and do not arm, change mode, move gimbal or execute guided actions by accident.

T12 - End-To-End Flight Smoke Test

Critical

Goal: Confirm the custom build is usable for a normal pilot workflow.

1. Connect vehicle, telemetry, camera and gimbal.
2. Verify pre-arm status, battery, GPS and link quality.
3. Arm using the approved pilot procedure.
4. Take off to a safe low altitude.
5. Hover and check HUD, map, video, gimbal, FOV and toolbar indicators.
6. Change to one approved flight mode and back.
7. Land and disarm.
8. Confirm logs, screenshots or screen recording are saved if required.

Expected result: no critical UI, telemetry, camera, gimbal or control regression appears during the basic flight workflow.

6. Defect Log

Use this table during the test session. Attach screenshots, video timestamps, QGroundControl logs and autopilot logs where possible.

Test ID	Pass / Fail	Issue Description	Evidence	Severity
T01				
T02				
T03				
T04				
T05				
T06				
T07				
T08				
T09				
T10				
T11				
T12				

7. Pilot Sign-Off

RECOMMENDED RESULT

Pass

Build is acceptable for the next test stage.

CONDITIONAL RESULT

Retest

Minor issues found. Fix and repeat affected tests.

BLOCKED RESULT

Fail

Safety, control, telemetry, video or gimbal issue found.

Test Pilot Name

Date:

Reviewer / Engineer Name

Date: